ANSWER KEY

English Test

1. C	21. D	41. D	61. D
2. F	22. J	42. F	62. F
3. A	23. D	43. D	63. C
4. J	24. J	44. G	64. F
5. D	25. A	45. C	65. B
6. H	26. F	46. J	66. G
7. B	27. B	47. C	67. D
8. J	28. H	48. H	68. G
9. D	29. B	49. C	69. B
10. F	30. H	50. F	70. J
11. A	31. B	51. B	71. B
12. G	32. H	52. H	72. J
13. D	33. B	53. B	73. A
14. F	34. F	54. H	74. J
15. B	35. B	55. A	75. D
16. H	36. F	56. G	
17. A	37. C	57. D	
18. G	38. J	58. G	
19. C	39. D	59. C	
20. G	40. G	60. H	

Mathematics Test

1. B	21. B	41. B
2. K	22. F	42. H
3. C	23. B	43. A
4. H	24. J	44. F
5. C	25. A	45. B
6. J	26. G	46. K
7. B	27. B	47. A
8. H	28. G	48. H
9. D	29. E	49. D
10. K	30. G	50. G
11. C	31. C	51. D
12. K	32. F	52. J
13. A	33. C	53. C
14. F	34. G	54. H
15. B	35. B	55. C
16. H	36. K	56. K
17. B	37. A	57. A
18. H	38. F	58. F
19. E	39. A	59. E
20. G	40. G	60. H

Reading Test

1. A	21. D
2. H	22. H
3. B	23. A
4. J	24. H
5. C	25. B
6. F	26. H
7. D	27. D
8. G	28. J
9. D	29. A
10. J	30. G
11. B	31. A
12. F	32. H
13. D	33. C
14. H	34. F
15. B	35. A
16. G	36. J
17. C	37. D
18. J	38. H
19. D	39. C
20. F	40. F

Science Reasoning Test

1. C	21. B
2. H	22. H
3. C	23. C
4. G	24. G
5. A	25. C
6. H	26. F
7. C	27. A
8. H	28. G
9. A	29. B
10. F	30. F
11. D	31. D
12. G	32. F
13. A	33. C
14. G	34. G
15. B	35. A
16. J	36. G
17. D	37. C
18. G	38. H
19. B	39. D
20. G	40. J

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SCORING GUIDE

Your final reported score is your COMPOSITE SCORE. Your COMPOSITE SCORE is the average of all of your SCALE SCORES.

Your SCALE SCORES for the four multiple-choice sections are derived from the Scoring Table on the next page. Use your RAW SCORE, or the number of questions that you answered correctly for each section, to determine your SCALE SCORE. If you got a RAW SCORE of 60 on the English test, for example, you correctly answered 60 out of 75 questions.

Step 1 Determine your RAW SCORE for each of the four multiple-choice sections:

English _____

Mathematics _____

Reading _____

Science Reasoning _____

The following Raw Score Table shows the total possible points for each section.

RAW SCORE TABLE	
KNOWLEDGE AND SKILL AREAS	**RAW SCORES**
ENGLISH	75
MATHEMATICS	60
READING	40
SCIENCE REASONING	40
WRITING	12

Multiple-Choice Scoring Worksheet

Step 2 Determine your SCALE SCORE for each of the four multiple-choice sections using the following Scoring Worksheet. Each SCALE SCORE should be rounded to the nearest number according to normal rules. For example, 31.2 ≈ 31 and 31.5 ≈ 32. If you answered 61 questions correctly on the English section, for example, your SCALE SCORE would be 29.

English	_____ RAW SCORE	× **36** =	_____	÷ **75** =	_____	**− 2**	(*correction factor)
					_____ SCALE SCORE		
Mathematics	_____ RAW SCORE	× **36** =	_____	÷ **60** =	_____	**+ 1**	(*correction factor)
					_____ SCALE SCORE		
Reading	_____ RAW SCORE	× **36** =	_____	÷ **40** =	_____	**+ 2**	(*correction factor)
					_____ SCALE SCORE		
Science Reasoning	_____ RAW SCORE	× **36** =	_____	÷ **40** =	_____	**+ 1.5**	(*correction factor)
					_____ SCALE SCORE		

*The correction factor is an approximation based on the average from several recent ACT tests. It is most valid for scores in the middle 50% (approximately 16–24 scale composite score) of the scoring range.

The scores are all approximate. Actual ACT scoring scales vary from one administration to the next based upon several factors.

If you take the optional Writing Test, you will need to combine your English and Writing scores to obtain your final COMPOSITE SCORE. Once you have determined a score for your essay out of 12 possible points, you will need to determine your ENGLISH/WRITING SCALE SCORE, using both your ENGLISH SCALE SCORE and your WRITING TEST SCORE. The combination of the two scores will give you an ENGLISH/WRITING SCALE SCORE, from 1 to 36, that will be used to determine your COMPOSITE SCORE mentioned earlier.

Using the English/Writing Scoring Table, find your ENGLISH SCALE SCORE on the left or right hand side of the table and your WRITING TEST SCORE on the top of the table. Follow your ENGLISH SCALE SCORE over and your WRITING TEST SCORE down until the two columns meet at a number. This number is your ENGLISH/WRITING SCALE SCORE and will be used to determine your COMPOSITE SCORE.

Step 3 Determine your ENGLISH/WRITING SCALE SCORE using the English/Writing Scoring Table on the following page:

English _____

Writing _____

English/Writing _____

ENGLISH/WRITING SCORING TABLE

ENGLISH SCALE SCORE	WRITING TEST SCORE											ENGLISH SCALE SCORE
	2	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	
36	26	27	28	29	30	31	32	33	34	32	36	**36**
35	26	27	28	29	30	31	31	32	33	34	35	**35**
34	25	26	27	28	29	30	31	32	33	34	35	**34**
33	24	25	26	27	28	29	30	31	32	33	34	**33**
32	24	25	25	26	27	28	29	30	31	32	33	**32**
31	23	24	25	26	27	28	29	30	30	31	32	**31**
30	22	23	24	25	26	27	28	29	30	31	32	**30**
29	21	22	23	24	25	26	27	28	29	30	31	**29**
28	21	22	23	24	24	25	26	27	28	29	30	**28**
27	20	21	22	23	24	25	26	27	28	28	29	**27**
26	19	20	21	22	23	24	25	26	27	28	29	**26**
25	18	19	20	21	22	23	24	25	26	27	28	**25**
24	18	19	20	21	22	23	23	24	25	26	27	**24**
23	17	18	19	20	21	22	23	24	25	26	27	**23**
22	16	17	18	19	20	21	22	23	24	25	26	**22**
21	16	17	17	18	19	20	21	22	23	24	25	**21**
20	15	16	17	18	19	20	21	21	22	23	24	**20**
19	14	15	16	17	18	19	20	21	22	23	24	**19**
18	13	14	15	16	17	18	19	20	21	22	23	**18**
17	13	14	15	16	16	17	18	19	20	21	22	**17**
16	12	13	14	15	16	17	18	19	20	20	21	**16**
15	11	12	13	14	15	16	17	18	19	20	21	**15**
14	10	11	12	13	14	15	16	17	18	19	20	**14**
13	10	11	12	13	14	14	15	16	17	18	19	**13**
12	9	10	11	12	13	14	15	16	17	18	19	**12**
11	8	9	10	11	12	13	14	15	16	17	18	**11**
10	8	9	9	10	11	12	13	14	15	16	17	**10**
9	7	8	9	10	11	12	13	13	14	15	16	**9**
8	6	7	8	9	10	11	12	13	14	15	16	**8**
7	5	6	7	8	9	10	11	12	13	14	15	**7**
6	5	6	7	7	8	9	10	11	12	13	14	**6**
5	4	5	6	7	8	9	10	11	12	12	13	**5**
4	3	4	5	6	7	8	9	10	11	12	13	**4**
3	2	3	4	5	6	7	8	9	10	11	12	**3**
2	2	3	4	5	6	6	7	8	9	10	11	**2**
1	1	2	3	4	5	6	7	8	9	10	11	**1**

Step 4 Determine your COMPOSITE SCORE by finding the sum of all your SCALE SCORES for each of the four sections: English only (if you do not choose to take the optional Writing Test) *or* English/Writing (if you choose to take the optional Writing Test), Math, Reading, and Science Reasoning, and divide by 4 to find the average. Round your COMPOSITE SCORE according to normal rules. For example, 31.2 ≈ 31 and 31.5 ≈ 32.

_____	+	_____	+	_____	+	_____	=	_____
ENGLISH *OR* ENGLISH/WRITING SCALE SCORE		MATHEMATICS SCALE SCORE		READING SCALE SCORE		SCIENCE SCALE SCORE		SCALE SCORE TOTAL

_____	÷ 4 =	_____
SCALE SCORE TOTAL		COMPOSITE SCORE

ANSWERS AND EXPLANATIONS

English Test Explanations

PASSAGE I

1. **The best answer is C.** The narrator was chosen "last spring," which was in the past. Answer choice A is incorrect because the moment the narrator is referring to is relative to the time the narrator wrote the passage, not another time in the past. Answer choices B and D can be eliminated because they are not past tense.

2. **The best answer is F.** The sentence appropriately uses the relative pronoun *who* to introduce the clause that modifies *couple*; this sentence is correct as it is written. The pronoun *who* also functions as the subject of the clause.

3. **The best answer is A.** The sentence represents a clear, complete thought that is grammatically correct. It is correct to begin a new sentence with *speaking*. You can eliminate the other answer choices because either they create incomplete sentences (B), or are otherwise grammatically incorrect.

4. **The best answer is J.** The sentence indicates that eating breakfast with the host family was a routine action in the past. Answer choice J is best because it includes *would*, which suggests repeated eating of breakfast with the couple. You can eliminate answer choice F because it is written in the present tense.

5. **The best answer is D.** The two clauses are unrelated and, therefore, you can connect the two separate ideas by using the word *then*. Answer choices B and C can be eliminated because they express a cause-and-effect relationship that does not fit with the sentence.

6. **The best answer is H.** This question asks you to place the apostrophe correctly in the underlined portion of the sentence. Answer choice H is best because the extended family is that of both parents together. This is made even clearer by the subject *they* in the following sentence.

7. **The best answer is B.** Answer choice B correctly uses the infinitive form *to tend* to explain why the host couple must travel. Choice C is incorrect because inflected forms like *tended* require an expressed subject. Choice D is incorrect because the *to* that follows is realized as a preposition linked to *travel*. The couple is not physically traveling to the *business*.

8. **The best answer is J.** This question tests your ability to spot relevance. Neither the underlined portion nor the information is answer choice G and H add anything relevant to the sentence. Therefore, it would be best to omit the underlined portion and simply end the sentence with "another."

9. **The best answer is D.** Logically, the question is asking whether the speaker would like to live with someone who speaks English or someone who does not speak English. The term *whether* only requires one side of a two-sided situation (e.g. "I don't know whether she has a roommate," not "I don't know whether she has a roommate or lives by herself.")

10. **The best answer is F.** The word *whom* is an object pronoun, meaning it will occur in object, not subject, position. In this case, it is the object of *find*. Remember, *whom* refers to Paolo, the logical direct object. This becomes clear if you reorder the clause: "I was surprised to *find Paolo* playing one of my favorite CDs on the stereo!"

11. **The best answer is A.** The essay maintains a positive, uplifting tone with regard to the speaker's friendship with Paolo. The other answer choices do not match the tone of the essay.

12. **The best answer is G.** The word *talk* takes a prepositional object starting with *about*; you cannot divide the phrase with a comma. Answer choice H is incorrect because modifiers like "my Brazilian friend" that come before proper names do not need a comma.

13. **The best answer is D.** The sentence is used to describe a faulty or incomplete first impression, then uses *but* to introduce a revision to it. Such clauses introduced by a subordinating conjunction are offset from the first clause of the sentence with a comma.

14. **The best answer is F.** The speaker is summarizing his trip into one important lesson. Answer choices G and H do not represent how meaningful the lesson is to the speaker and answer choice J is awkward as a modifier of *lesson*.

15. **The best answer is B.** The essay describes an enjoyable friendship between people of two different nationalities that begins in the context of a foreign culture. Answer choices A and C have a negative tone that does not match the rest of the essay. Answer choice D can be eliminated because it is not relevant to the passage.

PASSAGE II

16. **The best answer is H.** The clause subordinate to *apartment* is *I'm renting* and cannot be divided by a comma. The sentence up to the word *renting* is an adverbial, describing the location of the restaurant, so it must be followed by a comma.

17. **The best answer is A.** The paragraph is written in present tense. Answer choices B and C are past tense, and imply that the restaurant is not located there anymore. Answer choice D implies that the restaurant is not located there yet.

18. **The best answer is G.** The prepositional phrase introduced by *in* modifies *fountain* and cannot be divided by a comma. Answer choice J can be eliminated because a semicolon should be used to separate two independent clauses.

19. **The best answer is C.** Adjectives such as *sinful* modify nouns, such as *deliciousness*, which can be modified by prepositional phrases, such as *of Joe's Special Reuben*. The other answer choices incorrectly pair adjectives, adverbs, and nouns.

20. **The best answer is G.** This question tests your ability to recognize redundancy in a sentence. *Newcomers* implies people who have never seen *his creations*. The other answer choices are redundant.

21. **The best answer is D.** This question requires you to determine the correct punctuation. No punctuation is necessary between nouns like *sauerkraut* and gerunds like *spilling*, which modify them.

22. **The best answer is J.** Sentence 4 is about the menu in the window. It should be placed after the sentence that describes the menu. Placing it anywhere else in the paragraph would cause the paragraph not to make sense.

23. **The best answer is D.** This answer choice describes how the unwelcoming appearance of the restaurant does not reveal the truth about the delicious food. The other answer choices can be eliminated because they are not relevant to the topic of the paragraph.

24. **The best answer is J.** *Belies* is a transitive verb that takes what follows it as its direct object. A comma cannot be placed after *belies*; therefore, answer choices F and H can be eliminated. Answer choice G would create a sentence fragment, and can be eliminated.

25. **The best answer is A.** The verbs must agree with the subject *refrigerator*. Answer choice D is not appropriate when describing a single refrigerator case, and can be eliminated. Answer choice B does not make sense. Answer choice C is not inflected for tense.

26. **The best answer is F.** This sentence provides a logical transition between the previous sentence, which is about building repair, and the following sentence, which is about how the food is enough to sustain the charm of the restaurant.

27. **The best answer is B.** *Many* is the appropriate adjective for *"A lot of people"* from the previous sentence. It emphasizes the disparity between the narrator's knowledge of regular customers' faces and the sandwiches they eat. Answer choice A can be eliminated because while *much* can apply to mass nouns (e.g. much trash, much noise), it cannot be used with countable nouns. For those, *many* must be used (e.g. many people, many marbles). Answer choice C is an adverb and would, therefore, be ungrammatical. Answer choice D does not make sense in context.

28. **The best answer is H.** The phrase describes *the classics*, which by itself may be unclear. Answer choice G can be eliminated because the information is important to your knowledge of what *the classics* are. Answer choices F and J are not supported by the passage.

29. **The best answer is B.** The information is made clear without the use of excess commas or weak words like *makes* and *this place*. The other answer choices are too wordy or have awkward pauses.

30. **The best answer is H.** This question requires you to determine the correct punctuation. No punctuation is required between two noun phrases conjoined with *and*.

PASSAGE III

31. **The best answer is B.** The speaker is talking about Native Americans in the past; therefore, answer choices A, C, and D may be eliminated.

32. **The best answer is H.** A list of only two items does not need punctuation to separate them; therefore, the other answer choices may be eliminated.

33. **The best answer is B.** The speaker is talking about the ferries in past tense; therefore, the other answer choices may be eliminated.

34. **The best answer is F.** The phrase as it is written is clear and concise. Answer choice G is wordy and answer choices H and J do not make sense in the sentence.

35. **The best answer is B.** The underlined sentence explains that a potential bridge was reasonable because the Brooklyn Bridge was a success. The other answer choices are not supported by the passage.

36. **The best answer is F.** To "take action on" something is a common idiomatic expression; therefore, the other answer choices may be eliminated.

37. **The best answer is C.** The second clause describes something that stands in opposition to the first clause; that is, ferry service was resumed, *but* something caused it to stop again. The other answer choices would not link the two clauses correctly.

38. **The best answer is J.** The possessive form of "it" is "its," without the apostrophe; answer choice H may be eliminated. Here, "its" refers to "ferry service between peninsulas," which is singular; therefore, answer choice G may be eliminated.

39. **The best answer is D.** The sentence as it is written is a fragment. Simply eliminating "that" after "however" corrects this problem. The word "however," when used in the middle of a sentence must be offset with commas.

40. **The best answer is G.** The preceding sentence shows the significance of the bridge. Answer choices H and J can be eliminated because they are irrelevant to the topic of the essay. Answer choice F can be eliminated because the sentence does not explain how the bridge was built.

41. **The best answer is D.** The elements in the sentence are ordered logically. The other answer choices separate elements from each other with use of commas, which make the sentence less clear.

42. **The best answer is F.** The other answer choices would not make sense with the simple present form "is" used in the sentence.

43. **The best answer is D.** This question tests your ability to recognize redundancy. *Highway drivers* and *travelers* in this sentence essentially mean the same thing. The other answer choices are redundant.

44. **The best answer is G.** This answer choice does not fit with the topic of the essay, which is the Mackinac Bridge in the context of Michigan, not the context of the all the world's suspension bridges.

45. **The best answer is C.** This essay focuses on how the bridge came to be built. Answer choices A and B may be eliminated because the essay does not describe the entire process of building the Mackinac Bridge. Answer choice D may be eliminated because the topic of the essay is not specifically the reason for building the bridge.

PASSAGE IV

46. **The best answer is J.** The underlined sentence distracts from the intent of the paragraph, and should be omitted. The other answer choices also include information that is irrelevant to the topic of the essay.

47. **The best answer is C.** The narrator is speaking about his past. Answer choice A is ungrammatical and wordy. Answer C differs from B in that it omits whom, an unnecessary element. Answer D may be eliminated because the sentence would not make sense.

48. **The best answer is H.** This question requires you to determine the correct punctuation. No punctuation may come between a noun phrase (*box in his room*) and the gerund modifying it (*brimming*).

49. **The best answer is C.** The simple plural past tense is correct here. *No piece* is a singular subject, and cannot be referred to by answer choice A. Answer choice B would not make sense in the sentence.

50. **The best answer is F.** In Paragraph 2 the author analyzes the arguments in favor of electronic entertainment and then proposes his counterarguments. Answer choice G can be eliminated because it is irrelevant to the author's argument. Answer choice H is unsupported by the passage and answer choice J does not provide a logical transition from the previous paragraph.

51. **The best answer is B.** In the sentence, *but* does not correctly conjoin *dynamic* and *engaging*, which are closely related. The other answer choices are equivalent to the underlined sentences and are acceptable.

52. **The best answer is H.** When placed after Sentence 7, the sentence provides a logical transition to the subsequent paragraph. Answer choices F and

G can be eliminated because their placement would cause the paragraph to be illogical. The assertion in answer choice J is false, and may, therefore, be eliminated.

53. **The best answer is B.** This question tests your ability to recognize redundancy in a sentence. Answer choice B is the best answer because it avoids redundancy. Answer choices A, C, and D would introduce redundancy.

54. **The best answer is H.** The answer choice correctly modifies *weekend*. The narrator intends to describe card shows that occurred multiple times over multiple weekends. Answer choice F can be eliminated because it does not accurately convey what the narrator intends to describe.

55. **The best answer is A.** The sentence describes a habitual action in the past. Answer choice C can be eliminated because it is present tense. Answer choice B can be eliminated because *we* does not make any sense in the sentence.

56. **The best answer is G.** Adverbials like *too* are separated from the main clause by a comma. The gerund *encouraging* modifies *activity*, so *too* must be followed by a comma as well.

57. **The best answer is D.** This answer choice eliminates the redundancy. The word *screen* does not need to be repeated because the reader already knows that it is the video screen that silences the spectators.

58. **The best answer is G.** This answer choice is consistent with the narrator's use of the second person from the preceding sentence. Answer choices F, H, and J can be eliminated because they do not use second person.

59. **The best answer is C.** The best way to answer this question is to try the answer choices in the sentence. Answer choice C is the best because only adverbs modify adjectives.

60. **The best answer is H.** The narrator believes video games do not teach the skills card collecting teaches. Showing this in the passage indicates the author finds card collecting superior to electronic entertainment.

PASSAGE V

61. **The best answer is D.** This question requires you to determine the correct punctuation. In the sentence, *wait* takes a prepositional object headed by *until*, so punctuation must not separate them.

62. **The best answer is F.** The clause headed by *which* provides descriptive detail about why spyware is a threat. The remaining answer choices are not grammatical.

63. **The best answer is C.** The writer sets the computer virus in opposition to spyware, to show how the two are unlike each other. The other answer choices compare spyware with a computer virus, which is not the intent of the writer.

64. **The best answer is F.** The best way to answer this question is to try the answer choices in the sentence. Here, *often* coordinates well with *usually*. Answer choice H can be eliminated because *always* does not make sense with *but*. The other answer choices would not make sense in context.

65. **The best answer is B.** *However*, when used in the middle of a clause is set apart with commas. The other answer choices can be eliminated because they don't follow this rule.

66. **The best answer is G.** The paragraph describes some harmful spyware programs. This sentence effectively introduces the topic and links Paragraphs 2 and 3.

67. **The best answer is D.** This is clear, concise, and in active voice. The other answer choices are wordy.

68. **The best answer is G.** Answer choice G is the only choice that avoids redundancy. *Detection utilities* suggest that the utilities detect spyware. It is unnecessary to restate the fact that spyware is being detected.

69. **The best answer is B.** The verbs must agree with the plural subject *Detection utilities*. Answer choice B is the only answer in which *scan* and *remove* correspond with the plural subject.

70. **The best answer is J.** The writer is using present tense. Answer choices G may be eliminated because it is in past tense. *Computer* is the logical object of *protect*, and therefore, answer choices H and F can be eliminated.

71. **The best answer is B.** This is the only answer choice that provides a specific sound web surfing habit. Answer choice A is not specific enough to provide you with an example of a habit. Answer choices C and D are irrelevant to the argument the writer is making.

72. **The best answer is J.** Negative elements appear at the beginning of imperative clauses. *Never* is negative, and therefore, would fit best at the beginning of the sentence.

73. The best answer is A. The sentence begins with an "If" clause and ends with a clause describing the result. These two must be separated by a comma and the subject must be repeated. The other answer choices may, therefore, be eliminated.

74. The best answer is J. The sentence would provide a logical conclusion to the essay. Answer choices F, G, and H can be eliminated because if the sentence were placed in one of those paragraphs, it would not support the arguments it followed.

75. The best answer is D. Though the author might know how to protect computers from spyware, he makes no reference to its programming or the ethical issues that surround it.

Mathematics Test Explanations

1. **The correct answer is B.** To find the midpoint of two points, you can take the average of the x and y coordinates. If point X has coordinates $(-4,0)$ and point Y has coordinates $(0,-8)$, then the midpoint is:

$$\left(\frac{(-4+0)}{2}, \frac{(0+-8)}{2}\right)$$
$$\left(\frac{-4}{2}, \frac{-8}{2}\right)$$
$$(-2,-4)$$

2. **The correct answer is K.** To solve, use the Pythagorean Theorem ($c^2 = a^2 + b^2$). Using values from ΔMNO, you can set up the equation like this:

$$10^2 = 6^2 + NO^2$$
$$100 = 36 + NO^2$$
$$NO^2 = 100 - 36 = 64$$
$$NO = \sqrt{64} = 8$$

3. **The correct answer is C.** Since a distance in meters, M, can be approximated by multiplying a distance in yards, Y, by 1.0936, it follows that $M \approx Y(1.0936)$.

4. **The correct answer is H.** Because Seth has 4 plaid shirts and 5 solid-colored shirts, the total number of shirts is $4+5=9$. Of these 9 shirts, 4 are plaid. Thus the probability that a randomly selected shirt will be plaid is $\frac{4}{9}$.

5. **The correct answer is C.** Unless otherwise specified, average means "arithmetic mean," which is defined as the sum of a set of values divided by the number of values. Therefore, you can see that the average number of enrollments per day is $\frac{(17+19+23+14+25+28)}{6} = \frac{126}{6}$, or 21.

6. **The correct answer is J.** When parallel lines are cut by a transversal (such as segment PR in this problem), "alternate interior angles" are congruent. In this problem, $\angle y$ and $\angle 4$ are alternate interior angles, and so you can conclude that they are congruent.

7. **The correct answer is B.** The easiest way to solve this problem is to take 25% of the original price and deduct it from the original price. To find 25% of the sale price of the carton of paper, you would multiply \$27.00 by 25%, or 0.25. Therefore, the sale price of the carton of paper would be $\$27.00 - (\$27.00)(0.25) = \$27.00 - \$6.75$, or \$20.25.

8. **The correct answer is H.** To solve this problem, you must recognize that parallel lines always have the same slope. Remember that to find the slope of the line, you have to convert the equation $2x-3y=7$ into slope-intercept form ($y = mx + b$, where m is the slope):

$$2x - 3y - 2x = 7 - 2x$$
$$\frac{(-3y)}{3} = \frac{(-2x+7)}{(-3)}$$
$$y = \frac{-2x}{-3} + \frac{7}{-3}$$
$$y = \frac{2}{3}x - \frac{7}{3}$$

Thus the slope of this line, or any line parallel to it, is $\frac{2}{3}$.

9. **The correct answer is D.** In order to solve this problem, you must realize that if Andrew had \$28,000 remaining after paying 30% in taxes, then the \$28,000 constitutes $100\% - 30\%$ or 70% of the original prize, P. Therefore, $0.7P = 28{,}000$. Dividing by 0.7, you can conclude that the original cash value of the prize was $P = \frac{28{,}000}{0.7}$, or \$40,000.

10. **The correct answer is K.** To solve this problem, it is useful to assign values to the number of apples that Melissa and Marcia both possess. If Marcia has 10 apples, Melissa has $10 - 3 = 7$ apples. If Melissa gives 2 of her 7 apples to Marcia, Melissa is left with $7-2=5$ apples. When Marcia receives 2 more apples, she has $10+2=12$ apples. Since Marcia now has 12 apples and Melissa now has 5 apples, Melissa has $12-5=7$ fewer apples than Marcia.

11. **The correct answer is C.** The absolute value of a number is its distance from zero, regardless of whether it is positive or negative. Therefore, the value of $|5-9| = |-4| = 4$.

12. **The correct answer is K.** Since this problem requires you to multiply two binomials, you can utilize the FOIL (First, Outside, Inside, Last) method

to multiply the expressions.

$$\text{First: } (3m)(m^2) = 3m^3$$

$$\text{Outside: } (3m)(-n) = -3mn$$

$$\text{Inside: } (n)(m^2) = m^2n$$

$$\text{Last: } (n)(-n) = -n^2$$

Finally, add all these terms up to come up with your final answer. $(3m + n)(m^2 - n) = 3m^3 - 3mn + m^2n - n^2$.

13. The correct answer is A. To solve this problem, you must distribute and add like terms, as follows:

$$13 - 2(x + 5) =$$

$$13 - 2x - 10 = 2x + 3$$

14. The correct answer is F. Remember that the rule for exponents states that for base number b and exponents x and y, $(b^x)^y = b^{xy}$. Thus, when you apply the numbers from this problem, you find that $(n^7)^{11} = n^{(7)(11)} = n^{77}$.

15. The correct answer is B. To solve this problem, recognize that the repeating decimal has four places (0.3456), and that the fourth place is occupied by the number 6. Therefore, every place that is a multiple of 4 will be represented by the number 6. Since 217 is not divisible by 4, you know that the 217th digit cannot be 6; eliminate answer choice E. Because 216 is a multiple of 4, the 216th digit will be 6. Therefore, the 217th digit must be 3, the next digit in the repeating decimal.

16. The correct answer is H. If a square has side x, then its perimeter is $4x$; this is because a square is defined as a rectangle where all four sides are of equal length. Since the perimeter of the square is 48, then $48 = 4x$ and $x = \frac{48}{4} = 12$. Thus, the length of one side of the square is 12. The area of a square is defined as $(\text{side})^2$; therefore the area of this square is 12^2 or 144.

17. The correct answer is B. The easiest way to solve this problem is to remember that when two binomial expressions are multiplied, there is a predictable result. Take the following generalized example: $(x + a)(x - b) = x^2 - bx + ax - ab$. If $x^2 - bx + ax - ab = 0$, then the solutions to the equation are $x = -a$ and $x = b$. The product of the solutions is $-ab$. With this expression, $x^2+3x-21 = 0$, the product of the solutions $(-ab)$ is -21.

18. The correct answer is H. Remember that a difference of squares factors easily, such as: $a^2 - b^2 = (a + b)(a - b)$. Using the same technique, you can factor $a^{16} - 16$ into $(a^8 + 4)(a^8 - 4)$. The factor $(a^8 - 4)$ is another difference of squares, so it can be factored further into itself: $(a^8 - 4) = (a^4 + 2)(a^4 - 2)$. Of these factors, only $(a^4 + 2)$ is an answer choice.

19. The correct answer is E. Recall that dividing by a fraction is equivalent to multiplying by the reciprocal. When $\frac{1}{4}$ is substituted for n in the following expression, $\frac{2n - 5}{n}$, the result is:

$$\frac{2\left(\frac{1}{4}\right) - 5}{\frac{1}{4}}$$

$$= \left(\frac{2}{4} - 5\right)4$$

$$= 2 - 20 = -18$$

20. The correct answer is G. Since 90 minutes is equal to 1.5 hours, a proofreader who can read 40 pages in one hour can read (1.5)(40) or 60 pages in 1.5 hours.

21. The correct answer is B. The height, h, can be found using the Pythagorean Theorem ($c^2 = a^2 + b^2$):

$$5^2 = 3^2 + h^2$$

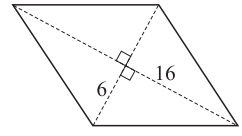
$$25 = 9 + h^2$$

$$h^2 = 16, \text{ or } h = 4.$$

Thus, when you multiply the base of the parallelogram by its height, the area of the parallelogram is $9 \times 4 = 36$.

22. The correct answer is F. For a certain quadratic equation $ax^2 + bx + c = 0$, if $x = \frac{a}{b}$ is a solution, then a possible factor would be $(bx - a)$. Since two solutions for $ax^2 + bx + c = 0$ are $x = \frac{3}{4}$ and $x = \frac{-2}{5}$, then possible factors are $(4x - 3)$ and $(5x + 2)$.

23. The correct answer is B. The diagonals of a rhombus intersect at their midpoints and form right angles as shown below.



Since the diagonals meet at their midpoints and form right angles, they form a right triangle with legs $\frac{12}{2} = 6$ and $\frac{32}{2} = 16$. To find the length of a side of the rhombus, you can simply use the Pythagorean Theorem and solve where the side of the rhombus, s, is the hypotenuse: $s^2 = 6^2 + 16^2 = 292$; s is approximately equal to 17.09.

24. The correct answer is J. If a rectangular parking lot has a length, l, that is 3 feet longer than its width, w, then $l = 3 + w$, or $w = l - 3$. The area of a rectangle is equal to its length times it width, or $A = lw$. Since the area of this parking lot is 550, $lw = 550$. Substituting $(l - 3)$ for

$$550 = l(l - 3) =$$

$$550 = l^2 - 3l$$

$$l^2 - 3l - 550 = 0.$$

To solve for l, factor the quadratic equation to get $(l + 22)(l - 25) = 0$, making $l = -22$ or $l = 25$. Since negative values for length do not make sense in this context, the length is 25.

25. The correct answer is A. To find the slope of the line between any two points (x_1, y_1) and (x_2, y_2), you can use the equation $\frac{(y_2 - y_1)}{(x_2 - x_1)}$. Therefore, when you have the points (3,7) and (4,−8) it follows that the slope of the line joining these points is $\frac{(-8 - 7)}{(4 - 3)} = \frac{-15}{1}$, or −15.

26. The correct answer is G. To find the solution set of $x + 2 > -4$, first solve for x by subtracting 2 from both sides. The result is $x > -6$. Thus the solution set is $\{x : x > -6)$.

27. The correct answer is B. To solve this problem, you need to know that the equation of a circle with center (h,k) and radius r is $(x - h)^2 + (y - k)^2 = r^2$. Therefore, the center of the circle in the problem, $(x - 3)^2 + (y + 3)^2 = 4$, is (3,−3).

28. The correct answer is G. To find the distance between two points (x_1, y_1) and (x_2, y_2), you can use the distance formula, which is $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$. The length of the line segment that has endpoints (−3,4) and (5,−6) will equal the distance between points (−3,4) and (5,−6). Therefore, $d =$

$$\begin{aligned}
&\sqrt{(5 - (-3))^2 + (-6 - 4)^2} \\
&= \sqrt{8^2 + 10^2} \\
&= \sqrt{64 + 100} \\
&= \sqrt{164} \\
&= \sqrt{(4)(41)} \\
&= \sqrt{4}\sqrt{41} \\
&= 2\sqrt{41}
\end{aligned}$$

29. The correct answer is E. The key to solving this problem is remembering that the triangle inequality states that no one side of a triangle can be greater than the sum of the other two sides. Thus the third side of the triangle in the problem cannot be greater than the sum of the other two sides, 4.7 and 9, which is 13.7. Of the answer choices, only 14 is too large to be a possible value for the third side of the triangle.

30. The correct answer is G. To solve this problem, recall that $\frac{n^x}{n^y} = n^{x-y}$. Since it is given in the problem that $\frac{n^x}{n^y} = n^2$, you can conclude that $n^{x-y} = n^2$ and thus $x - y = 2$.

31. The correct answer is C. To solve, convert the equation of the line to slope-intercept form ($y = mx + b$, where m is the slope and b is the y-intercept). If $3x + 5y = 8$, then:

$$3x + 5y - 3x = 8 - 3x$$

$$\frac{(5y)}{5} = \frac{(-3x + 8)}{5}$$

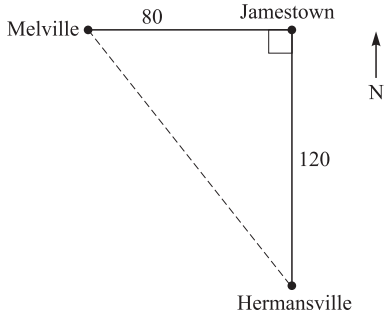
$$y = \frac{-3x}{5} + \frac{8}{5}$$

Since the equation $y = \frac{-3x}{5} + \frac{8}{5}$ is in slope-intercept form, the y-intercept is $\frac{8}{5}$.

32. The correct answer is F. To find the cost per ounce, first convert 3.4 pounds to ounces. Because there are 16 ounces in a pound, 3.4 pounds is $3.4(16) = 54.4$ ounces. To find cost per ounce, divide the cost in dollars by the number of ounces, or $\frac{\$4.95}{54.4}$ ounces = \$0.09 per ounce.

33. The correct answer is C. To solve, first square each fraction: $\left(\frac{1}{2}\right)^2 + \left(\frac{1}{3}\right)^2 + \left(\frac{1}{4}\right)^2 = \frac{1}{4} + \frac{1}{9} + \frac{1}{16}$. Remember that to be added, fractions must have a common denominator. In this case, since 4 is a factor of 16, the lowest common denominator is $(9)(16) = 144$. To convert fractions into different denominators, you must multiply the top and bottom of a fraction by the *same* number. If $\frac{1}{4}$ is multiplied by $\frac{36}{36}$, the result is $\frac{36}{144}$. Likewise, multiplying $\frac{1}{9}$ by $\frac{16}{16}$ yields $\frac{16}{144}$, and multiplying $\frac{1}{16}$ by $\frac{9}{9}$ yields $\frac{9}{144}$. Therefore $\frac{1}{4} + \frac{1}{9} + \frac{1}{16} = \frac{36}{144} + \frac{16}{144} + \frac{9}{144} = \frac{(36+16+9)}{144}$, or $\frac{61}{144}$.

34. The correct answer is G. The easiest way to solve this problem is to draw a picture similar to the one below.



Since the route heads straight north from Hermansville for 120 miles to Jamestown, and then straight west for 80 miles to Melville, the turn at Jamestown creates a right angle. If a straight, flat road existed between Hermansville and Melville, it would form the hypotenuse of a right triangle with legs 80 and 120. Using the Pythagorean Theorem ($c^2 = a^2 + b^2$), you can see that the distance of this straight route from Hermansville to Melville would be:

$$\sqrt{(120^2 + 80^2)}$$

$$= \sqrt{(14{,}400 + 6{,}400)}$$

$$= \sqrt{20{,}800} \approx 144$$

35. The correct answer is B. To solve this problem, calculate the volume of the aquarium and divide by 2. Since volume is equivalent to length × width *x* height, the volume is 30 × 16 × 12, or 5,760 cubic inches of water. Dividing by two, you see that half of the tank would be 2,880 cubic inches of water.

36. The correct answer is K. To solve this problem, you would multiply the number of possibilities in each officer position. Since the league selects its 4 officers by first selecting the president, then the vice president, then the secretary, then the treasurer, there are 40 possibilities for president, 39 possibilities for vice president, 38 possibilities for secretary, and 37 possibilities for treasurer. The total number of different possibilities for the election is therefore 40 × 39 × 38 × 37.

37. The correct answer is A. Because there is a right angle at *S*, the point *T* will lie along the line through *S* that is perpendicular to the segment *RS*. To solve this problem, find the equation for the line through *S* that is perpendicular to the segment *RS* and try each answer choice to find one that lies on the line. Since the line is perpendicular to segment *RS*, it will have a slope that is the opposite reciprocal of the slope of *RS*. Since slope is rise/run, the slope of *RS* is $\frac{(3-2)}{(6-2)} = \frac{1}{4}$. The slope of a line perpendicular to that is -4. Because a point and the slope of the line are known, the point-slope form of the equation can be utilized. A line through point $(h, -k)$ with slope m has equation $y - k = m(x - h)$. Thus the line through S $(6,-3)$ that is perpendicular to the segment *RS* has equation $y - 3 = -4(x - 6)$. Distributing and adding like terms, the result is $y = -4x + 27$. Of the answer choices, only the point (5,7) falls on the line.

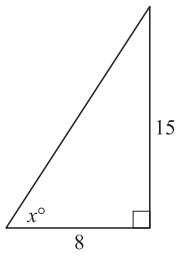
38. The correct answer is F. To solve the equation $0.2(x - 2{,}700) = x$, first distribute:

$$0.2x - 540 = x$$

$$-540 = 0.8x$$

$$-675 = x$$

39. The correct answer is A. Given that $0° \leq x \leq 90°$ and that $\tan x = \frac{15}{8}$, x can be pictured in the right triangle below.

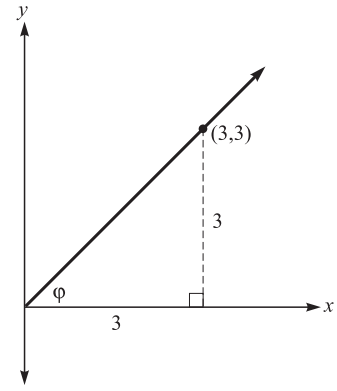


Because tangent is the ratio of the side opposite the angle to the side adjacent to the angle, the legs of the right triangle can be labeled as above. Cosine is the ratio of the side adjacent to the angle to the hypotenuse, which is not given. It is possible to eliminate answer choices in such a manner that it is not necessary to use the Pythagorean Theorem. Since the side adjacent to x is 8, the numerator in $\cos x$ will be 8, eliminating all but answers A and E. Since the legs of the triangle are 8 and 15, the hypotenuse will be longer than either, eliminating answer choice E. Thus $\cos x = \frac{8}{17}$.

40. The correct answer is G. Since the area of the square pool is given, you must find the area of the circle, with a radius of 10, and subtract the area of the pool. The area of a circle is equal to πr^2, where r is the radius. The area of this circle is $10^2\pi = 100\pi \approx 314$ square feet. Thus the area of the enclosure is approximately $314 - 81 = 233$ square feet.

41. The correct answer is B. Remember that all parallel lines have the same slope, so a line parallel to $y = 2x + 2$ will a slope of 2. A quick way to aid you in solving this problem would be to eliminate answer choices that do not have slope 2, so answer choices A and E can be immediately eliminated. Check the point (3,−4) in the remaining answer choices. The only choice that works is $y = 2x - 2$.

42. The correct answer is H. Tangent is the ratio of the side opposite to the side adjacent to an angle in a right triangle. Drawing a line that passes through (3,3) and is perpendicular to the x-axis creates a right triangle, as shown in the figure (see below).



Because point (3,3) is given, both legs of the right triangle have a length of 3. Thus $\tan \varphi = \frac{3}{3} = 1$.

43. The correct answer is A. To solve, calculate the result for each operation and select the smallest result.

Answer Choice A: $\frac{2}{3} + (-3) = -2\frac{1}{3}$

Answer Choice B: $\frac{2}{3} - (-3) = 3\frac{2}{3}$

Answer Choice C: $\frac{2}{3} \times -3 = -2$

Answer Choice D: $\frac{2/3}{-3} = \frac{-2}{9}$

Answer Choice E: $\frac{[2/3 + (-3)]}{2} = \frac{-7}{6}$

The smallest result is $-2\frac{1}{3}$, which was obtained by adding.

44. The correct answer is F. To simplify calculations, you can multiply the entire equation by 12 to obtain whole numbers and get $4b + 24 = 3$. Subtracting 24 from both sides yields $4b = -21$. Dividing by 4 yields $b = -\frac{21}{4}$, which is a little less than −5. Thus the correct answer will lie between −4 and −6.

45. The correct answer is B. To find the solution set for $|3a - 2| \leq 7$, break it up into two separate inequalities: $3a - 2 \leq 7$ and $3a - 2 \geq -7$. Starting with $3a - 2 \leq 7$, solving for a yields $a \leq 3$. With $3a - 2 \geq -7$, solving for a yields $a \geq -\frac{5}{3}$. Thus a is between $-\frac{5}{3}$ and 3 inclusive.

46. The correct answer is K. Tangent is the ratio of the side opposite to the side adjacent to an angle in a right triangle. If the distance, in feet, to the cell phone tower is x, then $\tan 41° = \frac{200}{x}$, or $x = \frac{200}{\tan 41°}$. Since $\cot 41° = \frac{1}{\tan 41°}$, $x = \frac{200}{\tan 41°} = 200 \cot 41°$.

47. The correct answer is A. Since the area of a square is equal to the square of its sides, multiplying the sides by $\sqrt{3}$ will have the effect of multiplying the area by $(\sqrt{3})^2 = 3$.

48. The correct answer is H. In order to solve this problem, you must realize that since the volume of a cube is equal to the cube of its sides, multiplying the length of the sides by $\frac{1}{2}$ will have the effect of multiplying the volume by $\left(\frac{1}{2}\right)^3 = \frac{1}{8}$. The cube in this problem has a volume of 64, so if you halve the length of each side, new cube's volume will be $64\left(\frac{1}{8}\right) = 8$.

49. The correct answer is D. The area of a parallelogram is equal to base × height. In the figure, you can see that the base of the parallelogram is 7 and the height of the parallelogram is 9. Thus, the area of the parallelogram is $9 \times 7 = 63$.

50. The correct answer is G. In order for $8a^6b^3$ to be less than zero, either 8 or a^6 or b^3 must be less than zero. However, it is obvious that $8 > 0$ and any number taken to an even power is non-negative. Thus $b^3 < 0$ and in order for that to be true, $b < 0$. Of the answer choices, only $b > 0$ CANNOT be true.

51. The correct answer is D. Logarithms are used to indicate exponents of certain numbers called bases. By definition, $\log_a b = c$, if $a^c = b$. If $\log_4 x = 3$, then $x = 4^3$, or 64.

52. The correct answer is J. In order for a system of 2 linear equations to have no solutions, the graphs of the equations must be parallel. Parallel lines have the same slope. To find the equation whose graph is parallel to the line in the figure, you must find the slope of the line between the points (0,−4) and (3,−0). Since slope is $\frac{\text{rise}}{\text{run}}$, the slope is $\frac{4}{3}$. The only equation with the correct slope of $\frac{4}{3}$ is $y = \frac{4}{3}x + 2$.

53. The correct answer is C. Of the 80 marbles, only 8 end in 5. If the first marble is drawn and not replaced, there are 79 marbles left, 7 of which have a ones digit of 5. Thus the probability that the player will be a winner is $\frac{7}{79}$.

54. The correct answer is H. To solve this problem, remember that the formula for slope is equal $\frac{(y_2 - y_1)}{(x_2 - x_1)}$, where (x_1, y_1) and (x_2, y_2) are two given points on a line. The equation of the line that passes through the origin and the point (3,4) will have slope $\frac{(4-0)}{(3-0)} = \frac{4}{3}$. Since the line passes through the origin, the y-intercept is 0. Thus the correct equation is $y = \frac{4}{3}x$.

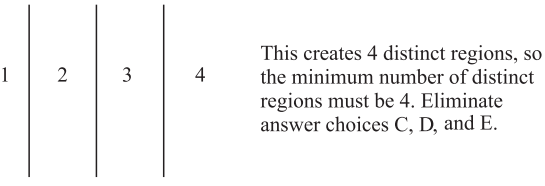
55. The correct answer is C. To solve this problem, you must remember that in an isosceles triangle, the base angles have the same measure. Since the sum of angles is 180° for all triangles, $180 = (a + 30) + 2(2a - 15)$. Distributing and adding like terms yields

$$180 = (a + 30) + 4a - 30$$
$$180 = 5a$$
$$a = 36$$

Since the base angles are equivalent to $2a - 15$, they equal $2(36) - 15 = 72 - 15$, or 57°.

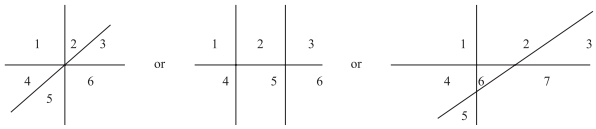
56. The correct answer is K. The smallest possible value will occur when it is negative. A negative product will result only when one of the numbers is positive and one is negative. The possible pairs are then −1 and 6, −2 and 5, −3 and 4, −4 and 3, −5 and 2, and −6 and 1. Of these pairs, the smallest product is $(-3)(4) = (-4)(3)$, or −12.

57. The correct answer is A. Start by drawing 3 parallel lines.



Now, try drawing 3 lines in other configurations, and you will see that there will always be either

6 or 7 regions:



Therefore, the correct answer is 4, 6, or 7 distinct regions, answer choice A.

58. The correct answer is F. Remember that the area of a parallelogram is equal to base × height. In this case, the base is $[3 - (-3)]$ or $[2 - (-4)]$, both of which equal 6, and the height is $(3 - (-5)) = 8$. Thus the area is $6 \times 8 = 48$. The area of triangle QRS, $\left[\frac{1}{2}(b)(h)\right]$, is half the area of the parallelogram, or 24.

59. The correct answer is E. Refer to the following chart to follow the patter of the sequence.

Term	1	2	3	4 ... n
Term	a	ab	ab^2	ab^4 ... ab^{n-1}

Since the power of b is one less than the number of term, the nth term will be ab^{n-1}. The 643rd term will then be $ab^{643-1} = ab^{642}$.

60. The correct answer is H. Since AB is longer than BC, there are only two possible configurations: B is between A and C or C is between A and B. In the case that B is between A and C, $AC = AB + BC = 19 + 13 = 32$. In the case that C is between A and B, $AC = AB - BC = 19 - 13 = 6$. Therefore, AC can be 6 and 32 only.

Reading Test Explanations

PASSAGE I

1. **The best answer is A.** Jake makes this statement in response to Katherine telling him that she sometimes has to "throw something together at the last minute." This suggests that it does not happen often and is unintentional. The other answer choices are not supported by the passage.

2. **The best answer is H.** During the conversation, Katherine says, "It's just that I work two back-to-back jobs every night," and later, "Things are just a little hard for us right now." You can infer that Katherine is working hard and barely getting by.

3. **The best answer is B.** When Jake says, "Katherine, Maxwell needs to have a healthy lunch," he indicates that Max is currently not eating well. This example has nothing to do with how well Katherine takes care of her son, but rather highlights one of the problems she is having in taking care of her son. The other answer choices all make reference to her trying in some way.

4. **The best answer is J.** The word *predicament* is used to indicate Jake's problematic situation with Max's lunches. Based on the context of the paragraph, his situation is somewhat difficult. Therefore, *predicament* most nearly means "challenge," which refers to a difficult task. The other answer choices are not supported by the context of the passage.

5. **The best answer is C.** Throughout the passage, Jake is never referred to as either Max's best friend or his brother, but Jake does know Max; therefore, A, B, and D should be eliminated. He is noted as being Max's camp counselor, and has a good influence on him by encouraging and allowing Max to consume healthier meals. Therefore, answer choice C is correct.

6. **The best answer is F.** At the end of the passage, Jake remarks that the club lasted for two summers, and then Max moved away. This best supports answer choice F. Although answer choice J may appear to be correct because it is implied that Max enjoyed the meetings of the Sandwich Club, the author never defines these meeting as Max's favorite part of camp.

7. **The best answer is D.** At several points in the passage, Katherine refers to how hard it is to find enough money for everything, and how much she has to work to make ends meet. This suggests that Katherine and Max have little money. The other answer choices are not supported by details in the passage.

8. **The best answer is G.** Whenever Jake brings up problems with Katherine, she seems genuinely concerned about Max's well-being. However, because she works so much and has so many other things to get done, her attention is often diverted away from Max. The other answer choices are not supported by the context of the passage.

9. **The best answer is D.** When Jake pulled Katherine aside to discuss the lunch issue, he indicated that Max's lunch is a concern of his, eliminating answer choices A and C. Although answer choice B may appear to be correct, the word "incident" implies a one-time occurrence. Max was consistently bringing unhealthy lunches to camp, therefore answer choice D, "problem," is correct.

10. **The best answer is J.** The passage states that Jake thought of the Sandwich Club because he needed to come up with a solution to help Max eat healthier lunches. The last paragraph also mentions the fact that once Max moved away, the Sandwich Club ended. These facts best support answer choice J. The other answer choices are beyond the scope of the passage.

PASSAGE II

11. **The best answer is B.** The passage states that "None but the fearless and inventive, the most resourceful and curious, would dare to undertake such a venture," which clearly suggests that this was a challenge. The passage later goes on to say that "All of the wonders of those states in the West are, in part, the result of this expedition." This implies that western expansion was important. Answer choice A may appear to be correct; however, the passages tells us that fur traders and trappers had traversed the wilderness to reach the west prior to Lewis and Clark's journey.

12. **The best answer is F.** The passage discusses many aspects of the Lewis and Clark expedition, and mentions several characteristic of the country, describing to the reader that the expedition led to "the unknown lands west of the Mississippi" and that "The path they were to carve out would be the first of its kind." Answer choice H may appear to be correct, however, the passage mentions that the men on the expedition were volunteers for "North Western Discovery," not the "Northwest Passage."

13. **The best answer is D.** The passage states that, "The settlers who came after Lewis and Clark went forward with blind-devotion knowing then that it could be done." This suggests that the Expedition gave other people confidence that they, too, could cross the United States because they knew that it had already been done by other travelers. Answer choice A is incorrect because the passage explicitly states that after Lewis and Clark's historic journey, many people traveled by land. Answer choices B and C are beyond the scope of the passage.

14. **The best answer is H.** The passage explicitly states that the Lewis and Clark expedition was intended to "find the fastest water route across North America." In addition, the passage states that previous to this journey, "virtually no one had attempted to cross the stretch of land between the mighty Mississippi and the vast Pacific Ocean using only water routes." Although Lewis and Clark did catalogue new species of animals on their journey—answer choice G—this was not their primary objective in traveling across the United States.

15. **The best answer is B.** The second paragraph states that, "Months later, in May, the party gathered in St. Lewis. The forty some men were to travel . . ." Answer choice B best fits the context of the paragraph. The previous statement does not support the idea that party means "a celebration," "a segment of the population," or a "meeting to discuss business," so answer choices A, C, and D can be eliminated.

16. **The best answer is G.** In the third paragraph, the men on the expedition are told to have written of their "health and high spirits." The statement is also made that the men were "all eager to explore," in spite of the potential dangers they faced in their long journey. This best supports answer choice G.

17. **The best answer is C.** It is reasonable to infer that the word "forge," as it is used in the passage – "the travelers continued to *forge* west in search of an efficient trade route using only the rivers"–refers to advancing on a path or journey, answer choice C. The other answer choices do not make sense within the context of the passage.

18. **The best answer is J.** The fourth paragraph mentions that "the Sioux and the Blackfeet tried to impede the group's progress on more than one occasion." This suggests that the Expedition would likely face danger in the form of conflict with indigenous people, answer choice J. The other answer choices are not supported by details in the passage.

19. **The best answer is D.** As stated in the passage, the men set out on three boats, "all of which were moved by sails, towropes, poles, or oars." These methods involve man power, wind power, or rowing power. Although the passage indicates that steam power eventually replaced the boats, nowhere does it mention that steam power was used on the Expedition.

20. **The best answer is F.** According to the passage, "After receiving wilderness training in Washington D.C., Meriwether Lewis set out . . ." This indicates that Lewis was the only member of the Expedition to have received wilderness training. All of the other answer choices are mentioned explicitly in the passage as having been done by all members of the Expedition.

PASSAGE III

21. **The best answer is D.** Although the passage discusses Porter's renowned ability to candidly address artists' works, illustrates the influence that several famous artists had on his works, and assesses his unusual methods of painting and critiquing artwork, none of these are the main focus of the passage. Therefore, answer choices A, B, and C can be eliminated. The passage gives a brief summary of Porter's life and discusses all of the above topics as points that shaped his career. This best supports answer choice D.

22. **The best answer is H.** At the end of the passage, the author states that it is sad that Porter "is still virtually unknown outside of art circles," and that "This remarkably insightful, articulate, creative individual needs to be discovered by the common man and revered for his continuing influence on the artists of today." Clearly, the author thinks very highly of Porter and his works and believes that he deserves to be honored ("revered") for his influence on today's artists. These details best support answer choice H. Answer choice F is incorrect because the author is clearly not "detached," or indifferent. Answer choice G is incorrect because the author is not merely "tolerant," or just able to withstand Porter. Answer choice J is incorrect because the author obviously does not "abhor," or hate, Porter and his works.

23. **The best answer is A.** As stated in the passage, "what made Porter so famous was his knack for

responding directly to an artist's work." Answer choices B and D are incorrect because the passage states that those are the things that Porter did NOT do, for he found criticisms based on those criteria to be insignificant and meaningless. Answer choice C is beyond the scope of the passage; Bernard Berenson's influence on Porter's art critiques is not discussed.

24. **The best answer is H.** Porter's personal philosophy regarding his paintings was that they should be "personal, emotional, and representative of its subject, while at the same time be boldly colorful, expressive, and generally abstract." This description of his works supports all of the answer choices except "trite," answer choice H, which means "ordinary or dull."

25. **The best answer is B.** The first paragraph of the passage simply tells how the author first came to meet Fairfield Porter, and provides a smooth transition into the life of Fairfield Porter. Answer choice A may appear to be correct; however, the author simply tells the reader how he came across Porter; there is no overview of what is to come in the passage. Answer choices C and D are not supported by information found in the passage.

26. **The best answer is H.** The passage states that, "Fairfield Porter, despite being remarkably intelligent, appeared to be lacking any natural artistic talents." Therefore, his high intelligence level was not correlated at all to his level of artistic ability. This best supports answer choice H. Porter's intelligence is not discussed in terms of the criteria listed in answer choices F, G, and J.

27. **The best answer is D.** The author states that, "This remarkably insightful, articulate, creative individual needs to be discovered by the common man and *revered* for his continuing influence on artists today." The author is clearly praising Porter and his continuing influence. It makes the most sense that the author believes that Porter should be "honored" for his continuing influence on artists today. The other answer choices do not make sense in the context of the paragraph.

28. **The best answer is J.** It is stated in the passage that Porter was primarily an art critic until "1961, when he decided to pursue a full time painting career." This was approximately 30 years after he returned from Italy, as he spent time between the years 1931 and 1932 in Italy learning to appreciate and critique artworks. The other answer choices are not supported by details in the passage.

29. **The best answer is A.** As stated in the passage, "Between the years 1931 and 1932, Fairfield spent the majority of his time in Italy learning to appreciate and critique the works of the great Renaissance painters. His training came from both . . ." This suggests that Porter was continuing his training as an art critic, answer choice A. The other answer choices are outside the scope of the passage.

30. **The best answer is G.** The passage preceding this phrase discusses Porter's fame as an acclaimed art critic. The passage then goes on to state that, "The other side of his fame, his uncommon approach to painting, is just as important to an understanding of Fairfield Porter's contributions to the world of art." This contrast indicates that Porter was famous both for his criticisms of art, as well as his artwork itself–answer choice G. Answer choice F may appear to be correct, however, the passage does not ever define Porter's level of fame as an artist versus his level of fame as an art critic.

PASSAGE IV

31. **The best answer is A.** The passage states that, "without dark matter, there are many cosmological phenomena that are difficult to explain." In the context of the sentence, it would make the most sense that without dark matter, there would be many "occurrences" or incidents that would be difficult to explain. Answer choice B may appear to be correct; however, the passage does not indicate the nature of the phenomena and whether or not they are problematic.

32. **The best answer is H.** The passage states that, "Dark matter . . . is not readily observable because it does not refract light or energy directly. Its existence can only be deduced because of the effect that it has on surrounding matter. In fact, some members of the scientific community have argued that dark matter does not actually exist." This best supports answer choice H. Answer choice F is incorrect because the evidence for the existence of dark matter is its effect on surrounding matter; this discussion of dark matter's effect on surrounding matter also eliminates answer choice J. Answer choice G is not supported by details in the passage.

33. **The best answer is C.** As stated in the passage, "It has been asserted that not only does dark matter exist, it may also be responsible for the Milky Way's unusual shape." The passage then goes on to discuss the way in which dark matter probably affects the shape of the Milky Way. Answer choice

B may appear to be correct; however, the passage does not indicate that the Magellanic clouds have enormous mass. In fact, the passage explicitly states that "The interaction referenced involves two *smaller* galaxies near the Milky Way, called Magellanic clouds, moving through an enormous amount of dark matter." Answer choices A and D are not supported by details in the passage.

34. The best answer is F. As stated in reference to Romanowsky's theory, "They point to the existence of several elliptical shaped galaxies surrounded by very little dark matter as evidence that dark matter is not, in fact, the cause of the warped galaxies." By showing galaxies that are similar in shape to the Milky Way but NOT surrounded by enormous amounts of dark matter, the theory illustrates that dark matter may not affect the shape and formation of galaxies, answer choice F. Answer choice G may appear to be correct; however, the passage explicitly states that Romanowsky's theory is not intended "to conclude that dark matter does not exist."

35. The best answer is A. The last paragraph states that "the movement of [the Magellanic] clouds through the dark matter seems to create a wake that enhances their gravitational influence on the Milky Way." Earlier in the paragraph, when discussing the Magellanic clouds, the passage states that when a cloud moves through dark matter, it "enhances the gravitational pull that the two Magellanic clouds could have on the Milky Way and other surrounding bodies." This best supports answer choice A. The other answer choices are not supported by details found in the passage.

36. The best answer is J. The third paragraph discusses the substance of Newton's hypothesis and the consequences that his hypothesis had on the existence of dark matter. When applying this hypothesis, it seems that dark matter must exist: "something that is not easily observed must be exerting the necessary force to create the warped shape of the galaxy." This best supports answer choice J. Answer choice F may appear to be correct; however, it is simply stated that Newtonian physics provide the strongest evidence for dark matter. It is not stated anywhere in the passage how dark matter was first discovered.

37. The best answer is D. The passage states that "they apparently believe that the results of their studies cast doubt on some of the *conventional* theories of galaxy formation and manipulation." Answer choices A and C do not make sense in the context of the sentence, because if a theory was "easily understood" or "strictly interpreted," there would not be a lot of room for doubt to be cast upon the theory. Answer choice B is incorrect because if a theory was "formally disputed," opposing viewpoints would already exist on that theory. That the result of these studies cast doubt on some "generally accepted" theories makes the most sense; therefore, answer choice D is correct.

38. The best answer is H. There is nothing in the passage to indicate with certainty that dark matter has no influence on surrounding celestial bodies. Although Aaron Romanowsky suggests that dark matter is not responsible for warped galaxies, there is no discussion in the passage to show that he believes dark matter has no influence whatsoever on galaxy shape or other cosmological phenomena. The other answer choices are all mentioned in the passage as scientific theories regarding dark matter.

39. The best answer is C. As stated in the passage, dark matter is "a substance that is not readily observable because it does not refract light or energy directly." This best supports answer choice C. The other answer choices are not supported by details found in the passage.

40. The best answer is F. The passage describes dark matter as surrounding and impacting galaxies composed of common matter, such as the Milky Way. This best supports answer choice F. Answer choice G is incorrect because whether dark matter truly exists is still a topic of debate among scientists. Answer choice H is incorrect because the passage does not provide the magnitude of the amount of dark matter in the universe. Answer choice J is incorrect because dark matter is not directly observable at all; its effect on galaxies is the only proof of its existence.

Science Reasoning Test Explanations

PASSAGE I

1. **The best answer is C.** Table 1 provides information on how many pinecones were produced at each of the six sites. When looking at the A4 and A5 column, the only sites at which A5 trees produced more pinecones per tree than A4 trees were S1, S2, S4, and S5. This best supports answer choice C.

2. **The best answer is H.** The results of Experiment 1 are shown in Table 1. According to these results, the only site at which A1 trees produced more pinecones than the other trees was at S4, answer choice H.

3. **The best answer is C.** According to Table 2, only S4 trees produced fewer pinecones than did the trees at S6. This suggests that A1 trees produced more pinecones at S6 than at S4, because A1 trees were planted with A2, A3, A4, and A5 trees in both experiments.

4. **The best answer is G.** Table 2 provides information on how many pinecones were produced per A1 tree at each of the six sites. To answer the question, you must look at which A1 tree produced the most pinecones. The A1 trees that were planted with seedlings from A2 at S3 produced 9.6 pinecones per tree, more than any other tree on the table. This information best supports answer choice G.

5. **The best answer is A.** To answer this question, you must read the procedures for both Experiment 1 and Experiment 2. In Experiment 2, "150 containers were prepared with 5 A1 seedlings and 5 seedlings from either A2, A3, A4, or A5." This says that, unlike Experiment 1, trees from more than one population were combined in Experiment 2. Therefore, answer choice A is correct.

6. **The best answer is H.** The procedure for Experiment 2 states, "150 containers were prepared with 5 A1 seedlings and 5 seedlings from either A2, A3, A4, or A5." Therefore, 10 seedlings were planted in each container. This data supports answer choice H.

PASSAGE II

7. **The best answer is C.** The point at hr = 16 for the "Ingredient A extended-release" line is at about 15 on the vertical axis. The point at hr = 16 for the "Ingredient B extended-release" line is at about 4 on the vertical axis. Therefore, the difference is about 11.

8. **The best answer is H.** This question can be answered by observing the trends in the Figures or simply thinking critically about how medication generally works. According to Figure 2, it takes a bit of time for the medication to start working, the symptoms decline for a period of time, then, presumably once the medication starts to wear off, the symptoms return. Figure 1 shows that symptom relief is inversely proportional to concentration of the medication in the body.

9. **The best answer is A.** According to Figure 1, the concentration of immediate-release Ingredient A increases most immediately following taking the pill. This is evident by the extended steep upward trend of the line.

10. **The best answer is F.** Table 1 shows that nearly as many subjects given the placebo (5%) reported feelings of weakness as did subjects given the drug (6%). The difference of only 1% shows that there is a high level of uncertainty over whether the drug actually caused the feelings of weakness.

11. **The best answer is D.** Figure 2 reflects the mean (average) symptom score of all the subjects given the drug. To find the 8-hour period of the graph where symptoms scores changed the least, find the 8-hour period of the curve over which slope changes the least. This occurs roughly between hr = 14 and hr = 22.

PASSAGE III

12. **The best answer is G.** By looking at Figure 1, you can determine the range in pressure at which each atmospheric layer can exist. Beginning with answer choice F, calculate the difference between the lowest pressure and the highest pressure at which the atmospheric layer can exist:

$$\text{Mesosphere: } 14.5 \text{ psi} - 7.5 \text{ psi} = 7 \text{ psi}$$

$$\text{Thermosphere: } 17.8 \text{ psi} - 10.3 \text{ psi} = 7.5 \text{ psi}$$

$$\text{Stratosphere: } 9.5 \text{ psi} - 3.8 \text{ psi} = 5.7 \text{ psi}$$

$$\text{Troposphere: } 5.0 \text{ psi} - 1.0 \text{ psi} = 4.0 \text{ psi}$$

The pressure in the thermosphere has the greatest range, answer choice G.

13. **The best answer is A.** To answer this question you need to look at the information on cumulus clouds in Figure 1. Figure 1 suggests that cumulus clouds form in the mesosphere. The mesosphere's

pressure ranges from 8 psi to 12 psi, and the average temperature for this range is 35°C. This information supports answer choice A.

14. **The best answer is G.** The key in Figure 1 includes two lines: temperature and pressure. As the pressure line is increasing (positive slope), the temperature line is decreasing (negative slope). This suggests that as pressure within the atmospheric layer increases, temperature decreases only, answer choice G.

15. **The best answer is B.** Answering this question requires you to look carefully at the range of temperature and pressure combinations at which clouds in the stratosphere form, as shown in Figure 1. Because the clouds here begin to form under 30°C and most are located below a pressure of 10 psi, this combination is correct. The other choices designate ranges of the atmosphere either above or below the stratosphere.

16. **The best answer is J.** At a pressure of 7 psi, Figure 1 shows that cloud formation will most likely occur in the stratosphere. According to Figure 1, clouds that form in the stratosphere are stratus clouds. This information best supports answer choice J.

PASSAGE IV

17. **The best answer is D.** Figure 1 shows the percent of fish that survive after exposure to water with bacteria present. The fish species with the highest percent (85%) of fish that survived after prolonged exposure to bacteria (light gray bar) was fish species E. Answer choice D is correct.

18. **The best answer is G.** To answer this question, you must look at the difference between the light gray and dark gray bars for each species in Figure 1. Answer choice J can be eliminated because the percent of fish that survived with and without exposure to bacteria are almost identical. Species C has the greatest difference in percent between the two, answer choice G.

19. **The best answer is B.** According to Table 1, fish that live in deep water without plants have a moderate ability to combat bacteria.

20. **The best answer is G.** By looking at Table 1, you can see that as you go down the column, the exposure to waterborne bacteria increase. Also as you go down the column, the relative ability to combat bacteria increases. The information supports answer choice G.

21. **The best answer is B.** According to Figure 1, Species A has the lowest percent of fish surviving to adulthood. Table 1 indicates that Species A fish live in shallow water with plants. Therefore, answer choice B is correct.

PASSAGE V

22. **The best answer is H.** This question asks you to identify the assumption that Paleontologist 1 must have made while discussing the finding of the fossilized bones. Answer choice F can be eliminated because the paleontologist refers to the velociraptor as a known dinosaur. Paleontologist 1 says, "The form and function of the velociraptor has been misunderstood until this important discovery." By stating that the dinosaur has been misunderstood, Paleontologist 1 is saying that it has been mischaracterized until now. This statement best supports answer choice H.

23. **The best answer is C.** Paleontologist 1 states, "You can see that there are cuts within the arm/wing bones of this dinosaur, indicating that it was caught while in flight." The paleontologist, then goes on to say, "Perhaps it was attempting an escape from a more predatory dinosaur." This suggests that while trying to escape from a larger predator, the predator caught the velociraptor, answer choice C.

24. **The best answer is G.** According to Paleontologist 2, "long arm bones are indicative of the dinosaur's ability to scavenge prey and fend off larger predators." It makes sense that Paleontologist 2 would conclude that, because the new dinosaur species had long arm bones, it could be a scavenger.

25. **The best answer is C.** According to Paleontologist 2, "The long arm bones are indicative of the dinosaur's ability to scavenge prey and fend off larger predators." This implies that the reason that the velociraptor had long arm bones was to scavenge for food.

26. **The best answer is F.** This question asks you to summarize the main idea of Paleontologist 1's viewpoint. Answer choices G and H can be eliminated because only Paleontologist 2 supports them. Paleontologist 1 says that "the velociraptor was definitely capable of flight" and "the velociraptor has been misunderstood until this important discovery." These two statements support answer choice F.

27. **The best answer is A.** Both Paleontologists mention the *Tyrannosaurus rex* as a possible threat either to the velociraptor's food source or to its very life. This best supports answer choice A. Answer choices B, C, and D do not support the views of both paleontologists, and therefore, can be eliminated.

28. **The best answer is G.** By reading the viewpoints of both Paleontologist 1 and Paleontologist 2, you can determine what they thought of the discovery of the velociraptor bones. Answer choices H and J can be eliminated because they are not supported by either passage. Both paleontologists extracted useful information from the discovery of the bones. Therefore, answer choice G is the best answer.

PASSAGE VI

29. **The best answer is B.** Figure 1 provides information on the compositions of mountain peaks. At a distance of 9 km along the mountain range, the peak composition is shown in the key as shale. This supports answer choice B.

30. **The best answer is F.** To determine the correct answer, you must look at the key in Figure 1, which shows the different composition of the mountain peaks. It defines limestone as "particles with diameters under 0.5 mm," shale as "composed mostly of particles with diameters under 175 mm," and slate as, "composed mostly of particles with diameters over 175 mm." The only difference between the different compositions is the size of the particle, answer choice F.

31. **The best answer is D.** As the peak section heights in Table 1 increase, the percentage of the year that peak section is exposed to wind also increases. Since a height of 5.5 m to 6.0 m is higher than the other values in the table, the percentage of the year that the peak section would be exposed to the wind should also be greater than the other values in the table. The only answer choice with a value greater than the other percentages in the table is answer choice D.

32. **The best answer is F.** Figure 2 shows that Peak D is exposed to wind erosion for a greater percentage of the year than Peak C. Table 1 suggests that the percentage of the peak exposed to wind is directly proportional to peak section height. Therefore, because Peak D is exposed to the wind for longer than Peak C is, Peak D must be taller than Peak C. This information best supports answer choice F.

33. **The best answer is C.** Table 1 shows that as the peak section height increases by equal increments, the percentage of the year that peak section is exposed to wind also increases by approximately the same amount. Therefore, the slope of the graph is positive and the graph is a straight line as shown in answer choice C.

34. **The best answer is G.** The passage states that "the peaks of mountains often lose sediment due to wind erosion." By losing sediment, the mountain peaks are losing mass. This supports answer choice G. Answer choice J can be eliminated because the information is irrelevant to the data presented in the passage.

PASSAGE VII

35. **The best answer is A.** According to the information in Table 2 and Table 3, the bacteria grew quicker when left under a lamp. This suggests that bacteria grow faster in warm environments, answer choice A. Answer choice C can be eliminated because it is irrelevant to the information presented in the passage.

36. **The best answer is G.** Table 3 provides information on the proportional growth of bacteria in four different environments. The moist side of Dish 2 produced 1.15 times the amount of bacteria in the control sample, which is more than the other three environments produced when compared to the control sample. This best supports answer choice G.

37. **The best answer is C.** Table 1 provides the results of Experiment 1. Answer choices A and D can be eliminated because they are not supported by the data. For both dishes, the bacteria growth was greater on the moist side. This information best supports answer choice C.

38. **The best answer is H.** More than one sample of bacteria was put into each Petri dish. To avoid skewed results, only one sample of bacteria should be placed in each Petri dish. Answer choice H is the best answer. Answer choice J can be eliminated because varying the size of the starting sample would not alter the results.

39. **The best answer is D.** Both Table 2 and Table 3 show that the bacteria growth rate is highest in

moist, warm places. The only answer choice that is both warm and moist is beneath a rock in a tropical forest. The other answer choices can be eliminated because they contain only one or none of the two conditions.

40. The best answer is J. According to the question, "bacteria double population size in short intervals." The only answer choice that includes recording data in short intervals is answer choice J.

Writing Test Explanation

Because grading the essay is subjective, we've chosen not to include any "graded" essays here. Your best bet is to have someone you trust, such as your personal tutor, read your essays and give you an honest critique. If you plan on grading your own essays, review the grading criteria and be as honest as possible regarding the structure, development, organization, technique, and appropriateness of your writing. Focus on your weak areas and continue to practice in order to improve your writing skills.

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